Lecture Notes on

Quantum Mechanics

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These notes cover quantum mechanics at the undergraduate level. The sources used are:

- Griffiths: *Introduction to Quantum Mechanics*

Potential sources to consult in the future:

- Sakurai: *Modern Quantum Mechanics*

Contents

1 The Schrödinger Equation

1.1 Wave Functions

Classically, we work with masses and attempt to solve for their equations of motions $x(t)$ using corresponding Newton's laws or energy equations. The quantum mechanics approach is different; the idea is to instead solve for a particle's wave function $\psi(t)$ and from there we can do anything. The goal then is to solve the Schrödinger equation

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar}{2m}\frac{\partial^2\psi}{\partial t^2} + V\psi$$

After solving the Schrödinger equation for some intial wave function $\psi(t)$ the equation actually gives us all future values of the wave function $\psi(t)$.
The statistical interpretation of the wave function goes as follows:

- Born's statistical interpretation of the wave function states that

  $$\int_b^a |\psi(x,t)|^2 dx$$

  is the probability of finding the particle at $a \leq x \leq b$ at time t. Note that while ψ is a complex valued function, $|\psi|^2 = \psi\psi^*$ where ψ^* is the complex conjugate of ψ is always real and nonnegative.

- An obvious consequence is this is that even if you know everything about a particle (ie; you know its wavefunction), you still cannot predict with certainty the outcome of an experiment to measure its exact position.

- Before measurement, a particle's position is not well defined. Only after measurement can we say the particle has a "position," and by doing so we collapse the particle's wavefunction to a dirac delta centered around the measured position. This is known as the **Copenhagen Interpretation**

- The integral over all possible positions must be equal to one, since the particle must be located somewhere.

  $$\int_{-\infty}^{\infty} |\psi(x,t)|^2 dx = 1$$

  If we have a wavefunction, we need to make sure its normalized for it to be physical. We can normalize it to obey the integral by multiplying it by a scale factor. A glance at the Schrödinger equation one can see that if $\psi(x,t)$ is a solution, then $A\psi(x,t)$ is also a solution for nonzero constant A.

- One can also show that the normalization integral does not change with time, so there is no need to constantly renormalized solutions as the particle evolves with time.

- If we interpret $|\psi(x,t)|^2$ to be the probability, then we can say the expectation value of position is

  $$\langle x\rangle = \int_{-\infty}^{\infty} x|\psi(x,t)|^2 dx$$

Note that the interpretation of this quantity is the expected value of the position from measuring particles with wave function ψ. Simply remeasuring the same particle over again will not work, since the first measurement will collapse the particle into some other wave function with exact position

- By differentiating the expected position with respect to time, one can obtain the expectation value of the velocity. It's customary for us to work using momentum so we have

$$\langle p \rangle = -i\hbar \int \left(\psi^* \frac{\partial \psi}{\partial x} \right) dx$$

- A more illustrative way to write the expectation values for position and momentum are:

$$\langle x \rangle = \int \psi^* \cdot x \cdot \psi dx$$

$$\langle p \rangle = -i\hbar \int \psi^* \cdot \left(\frac{\partial}{\partial x} \right) \cdot \psi dx$$

We say the the x operator "represents" position and the $-i\hbar\frac{\partial}{\partial x}$ operator "represents" momentum.

- All classical quantities can be represented as a function of position and angular momentum. Then the representation of a classical quantity $Q(x, p)$ can be written as

$$\langle Q(x, p) \rangle = \int \psi^* Q \left(x, -i\hbar \frac{\partial}{\partial x} \right) \psi dx.$$

- **Ehrenfest's Theorem** states that expectation values obey their classical laws. For example, the time derivative of momentum is

$$\frac{d\langle p \rangle}{dt} = \langle -\frac{\partial V}{\partial x} \rangle$$

Possible problems to include: 1.12, 1.17

1.2 Time Independent Schrödinger Equation

1.3 The Quantum Harmonic Oscillator

2 Dirac Notation

3 The Hydrogen Atom

4 Symmetries and Conservation Laws